

Chapter 1: Experimental chemistry

Measurement

Measure Time

Common units

→ 1 hour (h) = 60 minutes (min)

→ 1 minute (min) = 60 seconds (s)

SI unit → second (s)

Common apparatus → Digital stopwatch (measures 0.01s accuracy)

- e.g. 1.5s x 1.50s ✓ Decimal place very important!!!

Measure Mass

Common units

→ 1 tonne (t) = 1000 kilograms (kg)

→ 1 kilogram (kg) = 1000 grams (g)

→ 1 gram (g) = 1000 milligrams (mg)

SI unit → kilogram (kg)

Common apparatus → Electronic balance (measures 0.01g accuracy)

Measure Temperature

SI unit → Kelvin (K)

Common apparatus

→ Alcohol thermometer (measures 0.5°C accuracy)

→ Mercury-in-glass thermometer (measures 0.5°C accuracy)

Measure volume

Common units: ~~1m~~ $1\text{m}^3 = 1000\text{dm}^3 = 1\,000\,000\text{cm}^3$

$1\text{m}^3 = 1000\text{l} = 1\,000\,000\text{ml}$

SI unit → cubic metre (m^3)

Apparatus: Beaker, Measuring cylinder, Burette, Pipette (liquids)

Gas syringe (gases)

Common Apparatus to Measure Volumes in Liquids

1. Beaker

- Measures approximate volumes (e.g. 100cm^3 , 200cm^3)
- Not suitable for accurate measurements

2. Measuring cylinder

- Measures volumes to the nearest 0.5cm^3 (e.g. 36.0cm^3 , 62.5cm^3)
- Suitable for measuring the volume of "about $x\text{cm}^3$ of liquid"
- Not suitable for accurate measurements

3. Burette

- Measures volumes to the nearest 0.05cm^3 , up to 50.00cm^3 .
- Suitable for accurate measurements, usually seen by "exactly..."
→ Example: Exactly 15.40cm^3
- Not suitable for "about $x\text{cm}^3$..." or big volumes above 50cm^3 .

4. Pipette

- Measures ^{fixed} volumes very accurately. (e.g. 10.0cm^3 , 25.0cm^3 , 50.0cm^3)
- Only one measurement
- Only suitable for "exactly $10.0/25.0/50.0\text{cm}^3$ of..."

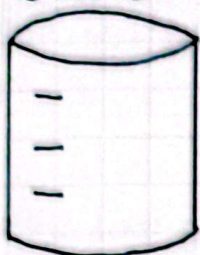
Common Apparatus to Measure Volumes of Gases

1. Gas Syringe

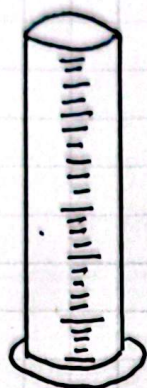
- A gas syringe is commonly used to collect and measure volumes of gases.
- As gas enters the syringe, the plunger is pushed outwards.

How to draw these apparatus to measure volumes

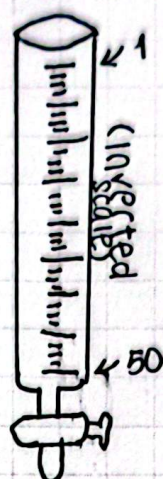
Beaker



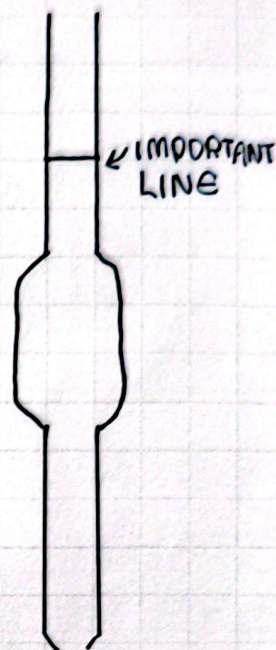
Measuring cylinder



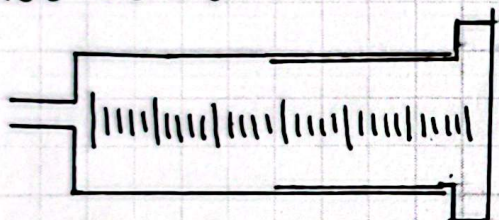
Burette



Pipette



Gas syringe



Collection of Gases

Methods to collect gases

a. Gas Syringe

b. Downward delivery, aka upwards displacement of air

c. Upward delivery, aka downwards displacement of air

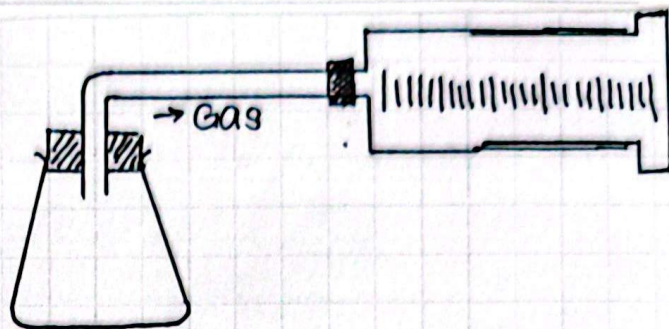
d. Displacement of water

★ The method used for collecting a gas depends on

- Solubility of the gas in water
- Density of the gas as compared to air

Gas Syringe

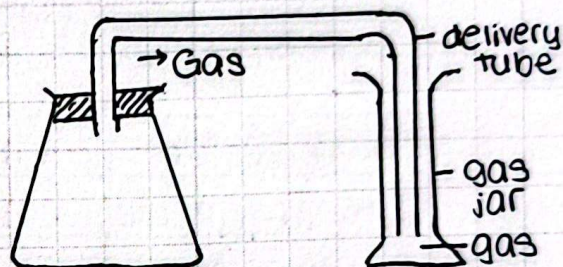
- This method can be used to collect and measure gases of all densities.



Downward Delivery or Upward displacement of air

- This method is used to collect gases that are denser than air.

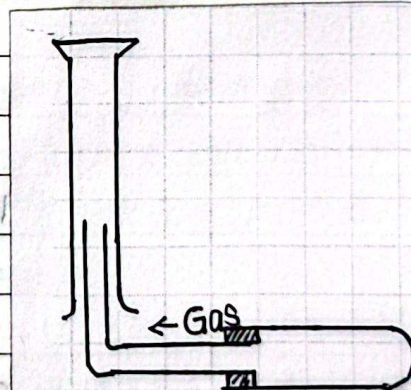
(Air $\rightarrow$ molecular mass of 28.8)



Upward Delivery or Downward displacement of air

- This method is used to collect gases that are less dense than air.

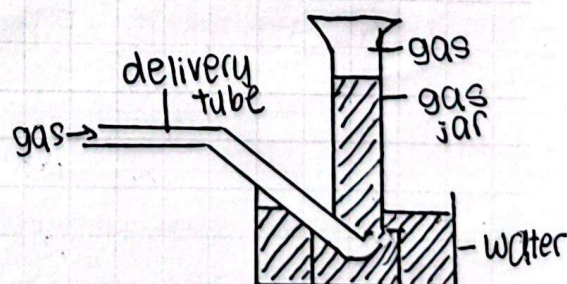
(Air $\rightarrow$ molecular mass of 28.8)



Displacement of water

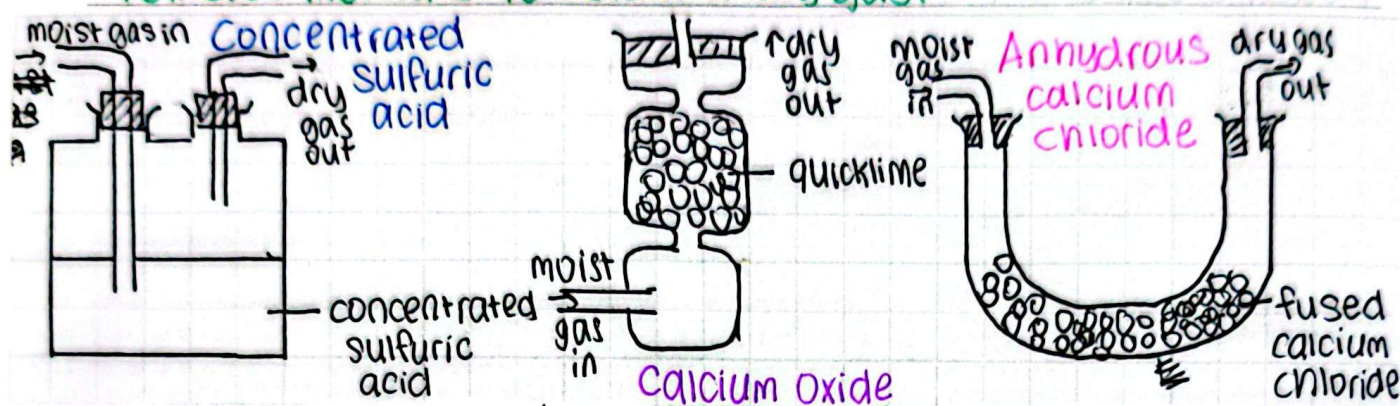
- This method is used to collect gases that are insoluble or slightly soluble in water.

- Not suitable for collecting dry sample of gas.



Drying Agents

- Often, the gas produced is contaminated with water vapour. The 'wet' gas is passed through a drying agent that removes moisture to obtain a 'dry' gas.



- This method cannot be used to dry alkaline gases as the acid neutralises alkaline gases.	- Quicklime = calcium oxide	- This method can be used to dry most gases except ammonia which reacts with carbon dioxide.
- Delivery tube of gas entering the acid is submerged in the acid.	- This method cannot be used to dry acidic gases as calcium oxide is a base that will neutralise acidic gases.	- Anhydrous Anhydrous = no water

Separation Techniques

Type of mixture		
Solid-Solid Mixture	Solid-Liquid Mixture	Liquid-Liquid Mixture
- Magnetic Attraction	- Filtration	- Separating funnel
- Sieving	- Evaporation to dryness	- Chromatography
- Using suitable solvents	- Crystallisation	- Fractional distillation
- Sublimation	- Simple distillation	

Separating Solid-Solid Mixtures

Using Magnetic Attraction

- A magnet can be used to separate ~~solids~~ magnetic solids ^{from} non-magnetic solids.
- e.g. iron filings and sulfur powder

Sieving

- A sieve can be used to separate solids with different particle sizes.

Using suitable solvents

- Liquids that dissolve solids are solvents.
- Solids that dissolve in the solvent are solutes.
- The ability of a solute to dissolve in a solvent is known as its solubility.

- A suitable solvent can be used to separate solid-solid mixtures in which only one of the solids is soluble in the solvent.

(Usually followed by filtration & evaporation to dryness)

Sublimation

★ Examples of solids that sublime:

- Iodine
- Dry ice aka solid carbon dioxide
- Naphthalene aka mothball
- Sublimation can be used to separate a substance that changes from solid to gaseous state directly.
- When heated, the solid sublimates into a gas. When cooled, the gas deposits into a solid, known as the sublimate.

Separating Solid-Liquid Mixtures

Filtration

- Filtration is used to separate an insoluble solid from a liquid.
- Filtrate → liquid that passes through the filter paper.
- Residue → solid that remains on the filter paper.

Decanting

- Used when the insoluble particles are large and heavy.
- The mixture is separated into 2 distinct layers, the layer on top is the filtrate and the bottom is the residue.
- The filtrate is poured off without pouring off the residue.

Evaporation to dryness

- Evaporation to dryness is used to obtain a soluble solid from its solution.

★ The solid must.

- Have a higher boiling point than the liquid.
- Be stable to heat.

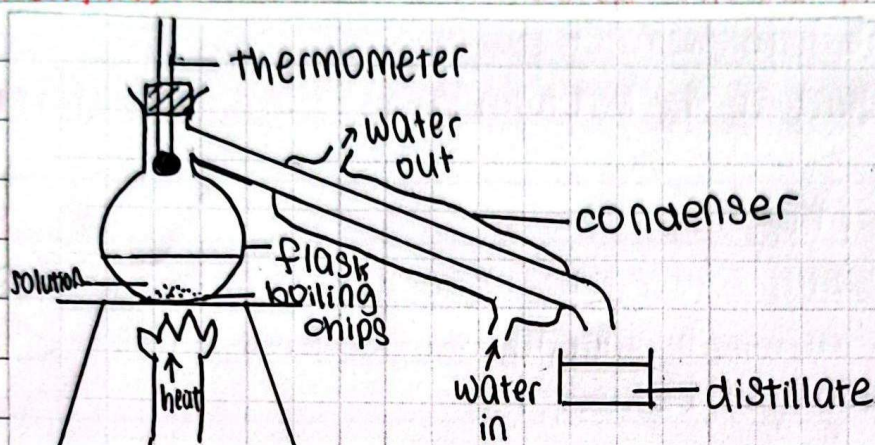
★ If the liquid in the mixture is flammable (e.g. alcohol), a water bath is used.

Crystallisation

- Crystallisation is used to obtain solids that are unstable to heat, e.g. sugar, copper(II) sulfate from its saturated solution. (They decompose upon heating)
- A saturated solution is a solution in which no more solute can be dissolved.
- The solution is usually heated to boil off some solvent to make the solution saturated. After cooling, crystals of solid appear.

Simple Distillation

- Simple distillation is used to obtain the solvent from a solution.
- The solid must have a way higher boiling point than the solvent (liquid).



- The solution is heated to its boiling point and the solvent changes into a vapour.
- The vapour is cooled by a condenser and changed back into a liquid.

Why are boiling chips added into the solution?

- To ensure smooth boiling.

Why is a thermometer placed near the opening of the ~~condenser~~ condenser?

- The thermometer measures the temperature of the vapour exiting the flask.

Why does cooling water enter the bottom of the condenser, not the top?

- The flow of water in the condenser is opposite to the flow of the vapour.
- This ensures that the bottom of the condenser is the coolest part to allow effective condensation of the vapour.

Separating Liquid-Liquid Mixtures

Separating Funnel

- A separating funnel is used to separate immiscible liquids.
- Immiscible liquids are liquids that do not mix at all and will separate into distinct layers.
- Densest liquid at the bottom; least dense liquid at the top

Chromatography

- Chromatography is used to separate a mixture of substances which have different solubilities in a given solvent.
- Chromatography can be used to:
 - determine if substance is pure
 - determine if substance(s) are soluble in water
 - separate the components in a sample

Procedure:

- Step 1: Draw a line on a piece of chromatography paper using pencil.
- Step 2: Put a tiny spot of sample on the pencil line.
- Step 3: Dip the chromatography paper into solvent (water).
The water should touch the paper but not the pencil line.
- Step 4: The solvent is absorbed by the paper and rises up.
- Step 5: Remove chromatography paper when the solvent has reached near to the top of the paper.

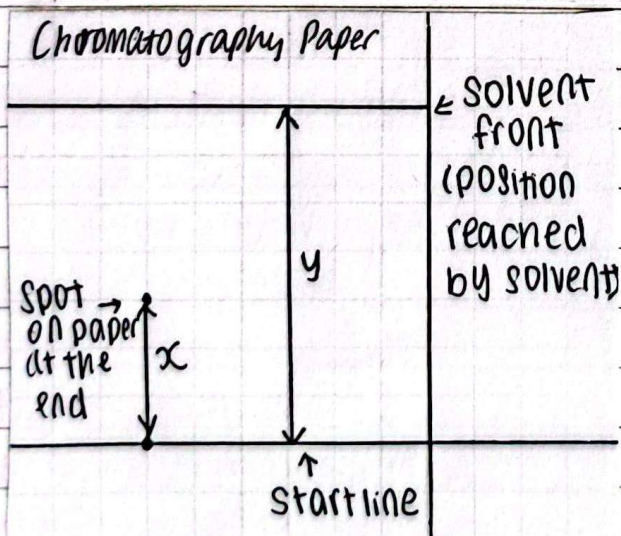
How to Interpret Chromatogram

- The number of spots indicate the number of components in the sample.
- If there are no spots, all the components in sample are insoluble.
- If there is one spot, the sample is a pure substance.
- If there is more than one spot, the sample is not pure i.e. mixture.
- A component that is more soluble in the solvent will travel faster and further across the paper.
- The difference in solubility allows the different components in a sample to be separated.
- Identical components produce spots at the same height.

Retention factor (R_f value)

- R_f values can be determined from the chromatogram.
- R_f value is the ratio of the distance travelled by solute compared to distance travelled by solvent.

$$\text{The retention factor (R}_f \text{ value)} = \frac{\text{Distance travelled by solute}}{\text{Distance travelled by solvent}}$$



$$R_f \text{ value} = \frac{x}{y} \text{ (always } < 1)$$

Comparing R_f values to known substances

- used to find the identity of unknown substances

R_f values of sugar:

Sucrose 0.20

Maltose 0.40

Glucose 0.57

Galactose 0.69

Fructose 0.86

Identifying colourless substances in chromatography

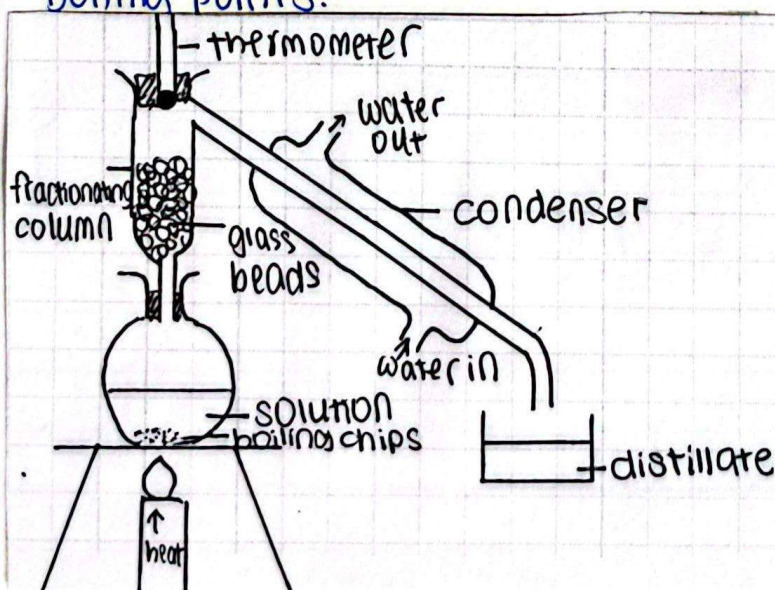
- A locating agent is applied on the chromatogram.
- The locating agent reacts with the substances to form coloured products and enable identification of substances.

Advantages of chromatography

- Only tiny amounts of substances are needed
- Results can be quickly obtained.

Fractional distillation

- Fractional distillation is used to separate miscible liquids (liquids that mix to form a solution) with different but close boiling points.

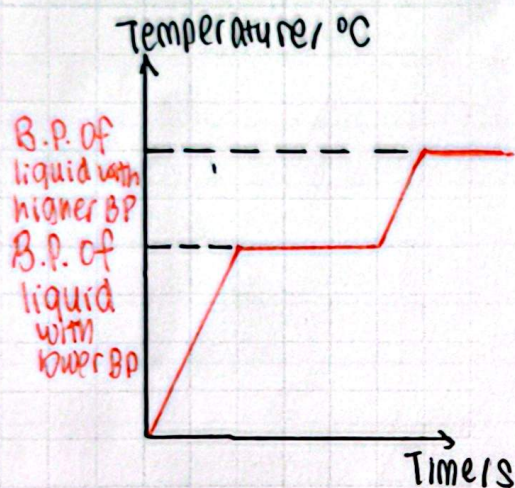


Function of glass beads:

- The fractionating column is packed with glass beads to provide a large surface area for condensation of vapour so components with different boiling points will leave the column at different times.
- Ensures effective separation of liquids.

- The liquid with lowest boiling point is distilled first while the liquid with highest boiling point is distilled last.

How to draw graph of fractional distillation



Collecting distillate

- The initial distillate collected is the substance with lower boiling point.
- To avoid collecting the other substance / liquid, the conical flask (receiver) must be removed when the thermometer reading rises ~~the~~ above the b.p. of the initial distillate.

Purity of substances

Pure substances

- A pure substance contains only one substance and is not mixed with other substances.
(i.e. one type of atom or molecules)
- A pure substance can be an element or a compound.

- For a substance to be considered pure, the substance must have a fixed boiling and melting point.

Impurities

- Impurities lower the melting point and cause the substance to melt over a range of temperatures.
- Impurities raise the boiling point and cause the substance to boil over a range of temperatures.

Determining purity of substance

- The purity of a substance can be determined by:

1. Checking the melting temperature of a solid

2. Checking the boiling temperature of a liquid

3. Performing chromatography

Mixtures

- Mixtures are when two ^{or more} substances are present and are not chemically joined together.

- A mixture is not a pure substance.

- Examples of mixtures:

- Solutions

- Suspensions